Master's Thesis

Microlocal Morse Theory and Truncated Microsupport
（超局所モース理論と打切り超局所台）

Toshihiro Oshiba

2026.3

Department of Mathematics, Graduate School of Science, Hokkaido University

Microlocal Morse Theory and Truncated Microsupport (超局所モース理論と打切り超局所台)

Toshihiro Oshiba

Table of Contents

1 Introduction

Microlocal Morse theory is a machinery to study the propagations of a cohomology of complexes of sheaves. More generally, this theory is considered as a part of microlocal sheaf theory, due to Kashiwara and Schapira. They defined the notion of microsupport of a complex of sheaves in the 1980's.

To explain more precisely, let us fix some notations. We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g., $\mathbf{Z}$ or a field). Let X be a topological space. We denote by $\mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$ (resp. $\mathsf{D}^{+}(\mathbf{k}_X)$) the bounded derived category (resp. bounded derived category from below) of sheaves of $\mathbf{k}$-modules on X.

If M is a real manifold of class C^∞ and F a complex of sheaves, then the microsupport $\mathrm{SS}(F)$ of F is a closed conic (i.e., stable under $\mathbf{R}^+$-action) involutive (i.e., coisotropic) subset of the cotangent bundle of M. The precise definition is given in Section 3, but roughly speaking, $\mathrm{SS}(F)$ describes the (co)directions of X in which the cohomology of F cannot be extended. This notion can be interpreted by analogy with Morse theory. Let φ be a real-valued function of class C^1. Then we can replace the notion of critical points of φ with the one with respect to F. That is to say, we can define a critical point with respect to F by the point p such that $d\varphi(p)$ belongs to $\mathrm{SS}(F)$. We can then obtain an analogous result of Morse theory in a framework of sheaf theory. In Section 5, we prove a truncated version of (a particular case of) the microlocal Morse lemma (cf. [GKS12, Proposition 1.8] or [KS90, Corollary 5.4.19]). Roughly speaking, this asserts that if there is no critical point with respect to a complex of sheaves, we have the natural isomorphism of the cohomologies of the sublevel sets of the manifold.

In 2003, Kashiwara, Monteiro-Fernandes, and Schapira introduced the notion of truncated microsupport in [KFS03a], in the course of the study of solutions of boundary value problems of $\mathcal{D}$-modules. The difference between microsupports and truncated microsupports is as follows. On the one hand, a point p in the cotangent bundle T^*M does not belong to the microsupport if and only if a certain kind of cohomologies of all the degrees vanish. On the other hand, the point p does not belong to the truncated microsupport if and only if the same kind of cohomologies of all the degrees less than a fixed number vanish. After they introduced truncated microsupports, some properties of truncated microsupports were shown. For example, a truncated microsupport does not satisfy the involutivity, but has a property called weak involutivity. This is shown also by Kashiwara, Monteiro-Fernandes, and Schapira in [KFS03b]. Some functorial properties of truncated microsupport are studied by several authors. See [KFS03a, FM07]. These authors studied truncated microsupports mainly in the context of analytic problems.

In this paper, in contrast to the previous research, we shall apply the truncated microsupport to the study of some geometrical problems. First we prove a truncated version of the noncharacteristic deformation lemma. Consider a Hausdorff space a family of open subsets of it. Assume that the family is "noncharacteristic" with respect to a complex of sheaves on the space. In this case, the original version of the theorem asserts that the global section on a member of the family is is isomorphic to the one on the union of the family. Note that this isomorphism is one between derived functors.

This lemma is one of the most fundamental results in microlocal sheaf theory, and used all along [KS90]. Thus a truncated version of this lemma is also the key result in this paper. However, one should notice that the original proof of the lemma cannot be applied to our truncated version. To prove the result, we make three modifications to the lemma. First, we add an assumption on a total space. That is, we assume that the space is not only Hausdorff, but also satisfies the condition that all the open subspaces are paracompact. Second, we weaken the hypothesis of vanishing of cohomologies. In the original lemma, we assume that the cohomologies of all degrees vanish. However, in our version, we assume that the cohomologies of not all degrees but degrees less than a certain number vanish. Third, we make an additional assumption on the family of deformation. Therefore, to make, in particular, the third assumption acceptable to the reader, we give several examples of families of deformation with figures. Then, in turn, we formulate truncated version (Theorem 5.3) of microlocal Morse lemma (Proposition 5.2), under the light of truncated microsupport. Note that we can also study the unique continuation property of cohomology groups of sheaves through the truncated microsupport. More precisely, if k is an integer and we consider the truncated microsupport $\mathrm{SS}_k(F)$ of a bounded complex of sheaves F, we can investigate whether the cohomology sheaves $H^j(F)$ propagate or not for $j < k$. Furthermore, we can see whether the cohomology sheaves $H^k(F)$ of the k-th degree enjoy the unique continuation property or not.

The paper is organized as follows. In Section 2, we review the notations and results from [KS90]. In Section 3, we define truncated microsupport of sheaves, due to [KFS03a]. In Section 4, we prove a truncated version of the noncharacteristic deformation lemma. In the proof, Lemma 4.4 plays a crucial role. However, since the proof of the lemma is long and rather technical, its proof will be given in Section 6. Finally in Section 5, we apply the result in Section 4 to the microlocal Morse theory.

2 Background Material

In this section, we recall some definitions and results of homological algebra and sheaf theory including the theory of derived categories from [KS90], following its notations with slight modifications.

Let X be a topological space, and Z a locally closed set of X. Let us denote by $i\colon Z \hookrightarrow X$ the inclusion. We use the following functors:

$$\mathrm{R}\Gamma_Z(\,\cdot\,) \coloneqq \mathrm{R}i_* i^!, \quad (\,\cdot\,)_Z \coloneqq \mathrm{R}i_! i^{-1}.$$

If there is no risk of confusion, we will write $\mathbf{k}_Z$ rather than $\mathbf{k}_{X,Z}$.

2.1 Geometrical Notions

Let M be a real manifold of class C^∞. We denote by $\tau_M\colon TM \to M$ the tangent bundle of M, and by $\pi_M\colon T^*M \to M$ the cotangent bundle of M. If there is no risk of confusion, we will write π rather than π_M. If $f\colon M \to N$ is a morphism of manifolds (i.e., a C^∞-map), then we have the natural diagrams.

$$\begin{tikzcd} TM \arrow[r, "f'"] \arrow[dr, "\tau_M"'] & M \times_N TN \arrow[r, "f_\tau"] \arrow[d] \arrow[dr, phantom, "\square"] & TN \arrow[d, "\tau_N"] \\ & M \arrow[r, "f"'] & N, \end{tikzcd} \qquad \begin{tikzcd} T^*M \arrow[dr, "\pi_M"'] & M \times_N T^*N \arrow[l, "f_d"'] \arrow[r, "f_\pi"] \arrow[d] \arrow[dr, phantom, "\square"] & T^*N \arrow[d, "\pi_N"] \\ & M \arrow[r, "f"'] & N. \end{tikzcd}$$

In these diagrams, f_τ and f_π denote the natural projections, f' is defined by $(x; v) \mapsto (x; df_x v)$, and f_d is induced by the transposition of f'.

2.2 Approximation by Inductive Limits

Proposition 2.1 ([KS90, Proposition 2.5.1]). *Let X be a topological space, Z a subspace, F a sheaf on X. Consider the canonical morphism:*

$$\psi\colon \varinjlim_U \Gamma(U; F) \to \Gamma(Z; F),$$

where U ranges through the family of open neighborhoods of Z in X. Then the following hold.

(i) The morphism ψ is injective.

(ii) Assume that X is Hausdorff and Z is compact. Then ψ is an isomorphism.

(iii) Assume that X is paracompact and Z is closed. Then ψ is an isomorphism.

Note that a paracompact space is, by definition, a Hausdorff space.

2.3 Complement of Homological Algebra

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML condition for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary. For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$E_n^\bullet \colon \cdots \to M_n^{j-1} \to M_n^j \to M_n^{j+1} \to \cdots , \tag{2.1}$$

and its projective limit

$$E_\infty^\bullet \colon \cdots \to M_\infty^{j-1} \to M_\infty^j \to M_\infty^{j+1} \to \cdots . \tag{2.2}$$

Denote by

$$\Phi_k \colon H^k(E_\infty^\bullet) \to \varprojlim_n H^k(E_n^\bullet) \tag{2.3}$$

the natural morphism.

Proposition 2.2 ([KS90, Proposition 1.12.4]). *Assume that for each $k \in \mathbf{Z}$, the system $\left(M_n^k\right)_n$ satisfies the ML condition. Then*

(a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,
(b) if moreover, for a given i the system $\left(H^{i-1}(E_n^\bullet)\right)_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 2.3 ([KS90, Proposition 1.12.6]). *Let $(X_s, \rho_{s,t} \colon X_t \to X_s)_{(s,(s \leqq t))}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps*

$$\lambda_s \colon X_s \to \varprojlim_{r<s} X_r$$

and

$$\mu_s \colon \varinjlim_{t>s} X_t \to X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0,s_1} $(s_0 \leqq s_1)$ are injective (resp. surjective).

2.4 ML Procedure for Sheaves

We use the following in the proof of Theorem 4.2.

Proposition 2.4 ([KS90, Proposition 2.7.1]). *Let X be a topological space and let $F \in \mathsf{D}^+(\mathbf{k}_X)$. Let $(U_n)_{n\in\mathbf{N}}$ be an increasing sequence of open subsets of X, $(Z_n)_{n\in\mathbf{N}}$ a decreasing sequence of closed subsets of X. Set $U := \bigcup_n U_n$, $Z := \bigcap_n Z_n$. Then the following hold.*

(i) For any j, the natural map $\varphi_j\colon H^j_Z(U;F) \to \varprojlim_n H^j_{Z_n}(U_n;F)$ is surjective.

(ii) Assume that for a given j, the projective system $(H^{j-1}_{Z_n}(U_n;F))_n$ satisfies the ML condition. Then φ_j is bijective.

(iii) Let now $(X_n)_{n\in\mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \mathrm{Int}(X_{n+1})$ for all n. Assume that for a given j, the projective system $\left(H^{j-1}(X_n;F)\right)_n$ satisfies the ML condition. Then the natural map $H^j(X;F) \to \varprojlim_n H^j(X_n;F)$ is bijective.

3 Truncated Microsupport

We give in this section the definition of a truncated microsupport and its functorial property under direct images of proper morphisms. Before that, we review the original definition of a microsupport.

Definition 3.1 ([KS90, Definition 5.1.2]). *Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$. The closed conic subset $\mathrm{SS}(F)$ of T^*X is defined by following condition: $p \notin \mathrm{SS}(F)$ if and only if there exists an open neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1-function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has*

$$\mathrm{R}\Gamma_{\{\varphi \geqq 0\}}(F)_x = 0. \tag{3.1}$$

Kashiwara, Monteiro-Fernandes, and Schapira modified this definition and gave the notion of truncated microsupport.

Definition 3.2 ([KFS03a, Definition 3.4]). *Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$, and fix an integer $k \in \mathbf{Z}$. The* truncated microsupport *$\mathrm{SS}_k(F)$ is the closed conic subset of T^*X defined by the following condition: $p \notin \mathrm{SS}_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1-function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has*

$$H^j_{\{\varphi \geqq 0\}}(F)_x = 0 \quad \textit{for any } j \leqq k. \tag{3.2}$$

For other equivalent conditions of the definition, see [KFS03a]. Truncated microsupports satisfy the following functorial property under proper direct images.

Proposition 3.3. *Let $f\colon X \to Y$ be a morphism of manifolds and let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$ such that f is proper on $\mathrm{supp}(F)$. Then for any $k \in \mathbf{Z}$,*

$$\mathrm{SS}_k(\mathrm{R}f_*F) \subset f_\pi f_d^{-1}(\mathrm{SS}_k(F)). \tag{3.3}$$

Moreover if f is a closed embedding, the inclusion above is an equality.

For other functorial properties, see [KFS03a, FM07].

4 Truncated Noncharacteristic Deformation Lemma

In this section, we prove a truncated version of the noncharacteristic deformation lemma. First we state the result. After giving several examples, we prove Theorem 4.2. In the course of the proof, we use the key lemma (Lemma 4.4). Since its proof is rather long, we postpone the proof to Section 6.

Before we state the truncated version, let us review the original version of the noncharacteristic deformation lemma.

Proposition 4.1 ([KS90, Proposition 2.7.2]). *Let X be a Hausdorff space. Let $F \in \mathsf{D}^+(\mathbf{k}_X)$, and let $(U_t)_{t\in\mathbf{R}}$ be a family of open subsets of X. We assume the following conditions.*

(i) $U_t = \bigcup_{s<t} U_s$, for all $t \in \mathbf{R}$.
(ii) For all pairs (s,t) with $s \leqq t$, the set $\overline{U_t - U_s} \cap \operatorname{supp} F$ is compact.
(iii) Setting $Z_s := \bigcap_{t>s} \overline{U_t - U_s}$, we have for all pairs (s,t) with $s \leqq t$ and all $x \in Z_s - U_t$,

$$(\mathrm{R}\Gamma_{X-U_t}(F))_x = 0.$$

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$\mathrm{R}\Gamma\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \xrightarrow{\sim} \mathrm{R}\Gamma\left(U_t; F\right).$$

Our result is as follows.

Theorem 4.2 (Truncated Noncharacteristic Deformation Lemma). *Let X be a Hausdorff space such that all the open subsets are paracompact. Let $F \in \mathsf{D}^+(\mathbf{k}_X)$, and let $(U_t)_{t\in\mathbf{R}}$ be a family of open subsets of X. Let $k \in \mathbf{Z}$. For $s \in \mathbf{R}$, set $Z_s := \bigcap_{t>s} \overline{U_t - U_s}$. We assume the following conditions.*

(i) $U_t = \bigcup_{s<t} U_s$, for all $t \in \mathbf{R}$.
(ii) For all pairs (s,t) with $s \leqq t$, the set $\overline{U_t - U_s} \cap \operatorname{supp} F$ is compact.
(iii) For all $s \in \mathbf{R}$ and all $x \in Z_s - U_s$, we have

$$(H^j_{X-U_s}(F))_x = 0 \quad \text{for all } j \leqq k.$$

(iv) For all pairs (s,t) with $s < t$, we have $Z_s \subset U_t$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \xrightarrow{\sim} H^j\left(U_t; F\right) \quad \text{for all } j < k,$$

and the monomorphisms

$$H^k\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \hookrightarrow H^k\left(U_t; F\right).$$

Before we prove Theorem 4.2, let us give several examples of families of deformation.

Example 4.3. The condition (iv) of Theorem 4.2 is not contained in the original version (Proposition 4.1). Therefore let us give an example of $(U_t)_{t\in\mathbf{R}}$ that does not satisfy the condition (iv).

(i) Let $X = \mathbf{R}^2$, and for a real number t define the function

$$\varphi_t(x) := \begin{cases} \max\left(0, t(2-|x|)\right) & (|x| < 2), \\ 0 & (|x| \geqq 2). \end{cases}$$

Then let us denote by U_t the open subset of X

$$U_t := \{(x, y) \in X;\ y < \varphi_t(x)\}\,.$$

This family $(U_t)_t$ does not satisfy the condition (iv). Indeed, if $0 < s < t$, we have

$$Z_s = \{(x, y) \in X;\ y = \varphi_s(x), |x| \leqq 2\}$$

and the point $(2, 0) \in Z_s$ is not contained in U_t. Moreover, let us define a locally closed subset Z of X by

$$Z := \left\{(x, y) \in X;\ (x+2)^2 + (y-1)^2 = 1\right\} \setminus \{(-1, 0)\}\,.$$

For this Z we denote by $\mathbf{k}_Z$ the sheaf on X as in Section 2. If $j \leqq 0$, we have $H^j_{X-U_t}(\mathbf{k}_Z)_x = 0$ for all $t \geqq s$ and $x \in Z_s - U_t$, but the morphism $H^0(U_1; \mathbf{k}_Z) = \mathbf{k} \to 0 = H^0(U_0; \mathbf{k}_Z)$ is not injective. This gives an example that satisfies the conditions (i)–(iii) but not (iv), which results in that the consequence of the proposition does not hold. Hence the condition (iv) is necessary.

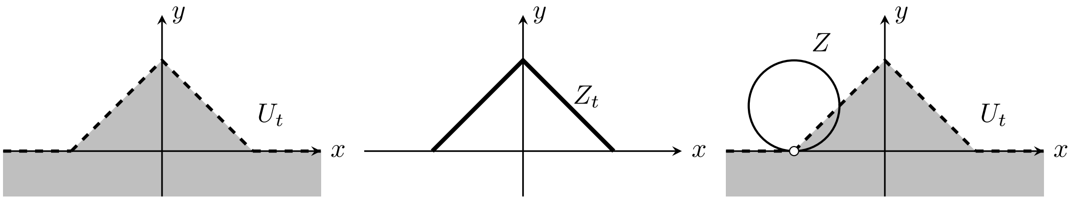

Fig.1　example (i)

Let us, in turn, give three examples of $(U_t)_{t \in \mathbf{R}}$ satisfying the condition (iv). In all the examples below, X is $\mathbf{R}^2$.

(ii) For a real number t, define the open set U_t of X by

$$U_t := \begin{cases} \left\{(x, y) \in X;\ x^2 + y^2 < t\right\} & (t > 0), \\ \varnothing & (t \leqq 0). \end{cases}$$

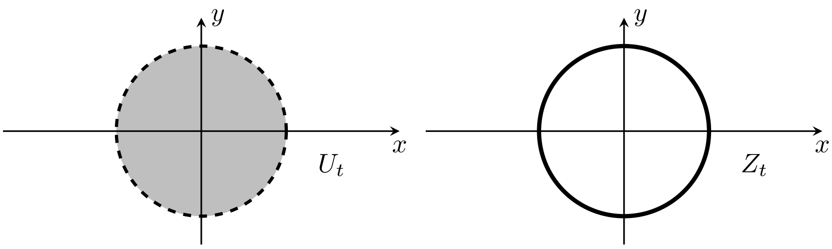

Fig.2　example (ii)

(iii) For a real number t, define the function $\varphi_t \colon \mathbf{R} \to \mathbf{R}$ as follows. If $t \leqq 0$, set $\varphi_t(x) \equiv 0$. If $t > 0$, set

$$\varphi_t(x) := \begin{cases} \sqrt{t^2 - x^2} & (|x| < t), \\ 0 & (|x| \geqq t). \end{cases}$$

Then define the open set U_t of X by

$$U_t \coloneqq \{(x,y) \in X;\ y < \varphi_t(x)\}\,.$$

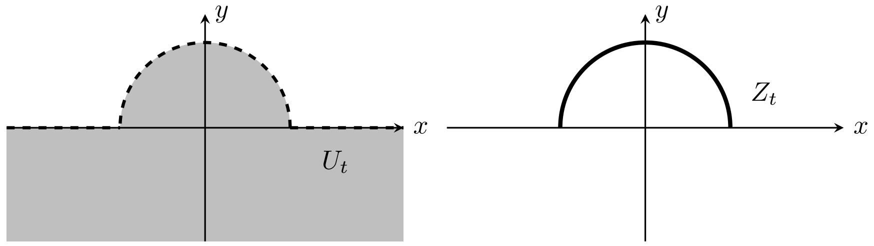

Fig.3　example (iii)

(iv) For a real number t, define the open set U_t of X as follows. If $t \leqq 0$, set

$$U_t \coloneqq \{(x,y) \in X;\ y < 0\}\,.$$

If $t > 0$, set

$$U_t \coloneqq \left\{ (x,y) \in X;\ \begin{array}{ll} \text{If } |x| < \frac{1}{t}, & y < t. \\ \text{Otherwise}, & y < \frac{1}{|x|}. \end{array} \right\}.$$

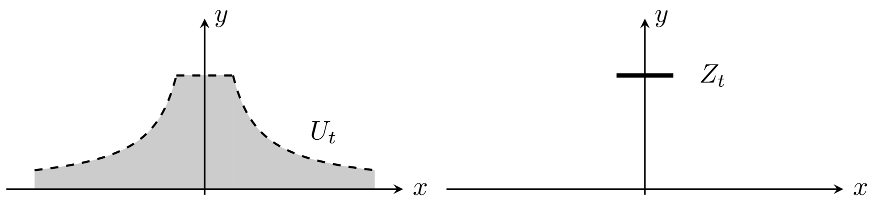

Fig.4　example (iv)

Proof of Theorem 4.2 Consider the following conditions for $j \in \mathbf{Z}$ and $s \in \mathbf{R}$:

$(a)^j_s$ $$\mu^j_s \colon \varinjlim_{t>s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F),$$

$(a')^k_s$ $$\mu^k_s \colon \varinjlim_{t>s} H^k(U_t; F) \hookrightarrow H^k(U_s; F),$$

$(b)^j_s$ $$\lambda^j_s \colon H^j(U_s; F) \xrightarrow{\sim} \varprojlim_{r<s} H^j(U_r; F).$$

Let us prove, for all $s \in \mathbf{R}$, $(a)^j_s$ for $j < k$ and $(a')^k_s$. We may assume that $\overline{U_t - U_s}$ is compact for any $s \leqq t$ by replacing X by $\operatorname{supp} F$. Consider the distinguished triangle

$$\mathrm{R}\Gamma_{X-U_s}(F) \to F \to \mathrm{R}\Gamma_{U_s}(F) \xrightarrow{+1}\,.$$

Let us endow with $Z_s \cup U_s$ the induced topology, and denote by $i_s \colon Z_s \cup U_s \hookrightarrow X$ the natural continuous map. Applying the functor

$$(\ \cdot\ )|_{Z_s \cup U_s} = i_s^{-1}$$

to the distinguished triangle above, we have

$$\mathrm{R}\Gamma_{X-U_s}(F)|_{Z_s\cup U_s} \to F|_{Z_s\cup U_s} \to \mathrm{R}\Gamma_{U_s}(F)|_{Z_s\cup U_s} \xrightarrow{+1} .$$

Again applying to this triangle the functor $\mathrm{R}\Gamma(Z_s \cup U_s;\ \cdot\ )$, we have

$$\mathrm{R}\Gamma(Z_s\cup U_s; \mathrm{R}\Gamma_{X-U_s}(F)|_{Z_s\cup U_s}) \to \mathrm{R}\Gamma(Z_s\cup U_s; F|_{Z_s\cup U_s}) \to \mathrm{R}\Gamma(U_s;F) \xrightarrow{+1} . \tag{4.1}$$

By the hypothesis (iii) and the definition of a local cohomology, we see that

$$H^j_{X-U_s}(F)\Big|_{Z_s\cup U_s} = 0 \quad \text{for any } j \leqq k.$$

Therefore we

$$H^j\left(Z_s\cup U_s; \mathrm{R}\Gamma_{X-U_s}(F)|_{Z_s\cup U_s}\right) = 0 \quad \text{for any } j \leqq k.$$

Thus we have from (4.1)

$$H^j(Z_s\cup U_s; F|_{Z_s\cup U_s}) \xrightarrow{\sim} H^j(U_s;F) \quad \text{for any } j<k, \tag{4.2}$$

$$H^k(Z_s\cup U_s; F|_{Z_s\cup U_s}) \hookrightarrow H^k(U_s;F). \tag{4.3}$$

Therefore, if we show Lemma 4.4 below, whose proof will be given in Section 6, we have from Lemma 4.4 and (4.2)

$$\begin{aligned}\varinjlim_{t>s} H^j(U_t;F) &\cong H^j(Z_s\cup U_s; F|_{Z_s\cup U_s})\\ &\xrightarrow{\sim} H^j(U_s;F)\end{aligned}$$

for any $j<k$, and $(a)^j_s$ follows. Similarly, we have from Lemma 4.4 and (4.3)

$$\begin{aligned}\varinjlim_{t>s} H^k(U_t;F) &\cong H^k(Z_s\cup U_s; F|_{Z_s\cup U_s})\\ &\hookrightarrow H^k(U_s;F)\end{aligned}$$

and $(a')^k_s$ follows.

Lemma 4.4. *For any $j \in \mathbf{Z}$ and $s \in \mathbf{R}$, there exists a natural isomorphism*

$$\varinjlim_{t>s} H^j(U_t;F) \cong H^j(Z_s\cup U_s; F|_{Z_s\cup U_s}).$$

We prove $(b)^j_s$ for all $j \in \mathbf{Z}$ and $s \in \mathbf{R}$ by induction on j. Since $F \in \mathsf{D}^+(\mathbf{k}_X)$, we have $(b)^j_s$ for $j \ll 0$ and $s \in \mathbf{R}$. Hence there exists an integer $j_0 < k$ such that $(b)^j_s$ holds for any $j < j_0$ and $s \in \mathbf{R}$. Then assume $(b)^j_s$ is proved for all $j < j_0$ and $s \in \mathbf{R}$. Since we have $(a)^j_s$ and $(b)^j_s$ for each $j < j_0$ and each $s \in \mathbf{R}$, we can apply Kashiwara's lemma (Proposition 2.3) and have

$$H^j(U_t;F) \xrightarrow{\sim} H^j(U_s;F) \qquad \text{for } j<j_0 \text{ and } s \leqq t.$$

Fix $s \in \mathbf{R}$ and consider the projective system $\left(H^{j_0-1}(U_{s-1/n})\right)_{n\in\mathbf{N}}$. This system satisfies ML condition. Therefore, by applying Proposition 2.4 (iii), we have

$$H^j(U_s;F) \xrightarrow{\sim} \varprojlim_n H^j\left(U_{s-\frac{1}{n}};F\right).$$

Since $(s-1/n)_n$ is a cofinal family of $(t)_{s>t}$, we have $(b)_s^{j_0}$. Then by induction on j, we have $(b)_s^j$ for $j<k$ and $s \in \mathbf{R}$. Again by applying Proposition 2.4 to the projective system $\left(H^j(U_n;F)\right)_n$, we obtain

$$\begin{aligned} H^j\left(\bigcup_{s\in\mathbf{R}} U_s;F\right) &\cong H^j\left(\bigcup_{n\in\mathbf{N}} U_n;F\right) \\ &\xrightarrow{\sim} \varprojlim_n H^j\left(U_n;F\right) \end{aligned}$$

and for each n, we have by Kashiwara's lemma (Proposition 2.3) that

$$H^j\left(U_n;F\right) \cong H^j(U_s;F)$$

for $n \geqq s$.

Finally we replace $(a)_s^j$ by $(a')_s^k$ for $j=k$, and obtain

$$H^k\left(\bigcup_{s\in\mathbf{R}} U_s;F\right) \hookrightarrow H^j\left(U_s;F\right).$$

This completes the proof. □

5 An Application to Microlocal Morse Theory

In this section, we shall apply the result in the previous section to a truncated version of a particular case of microlocal Morse lemma.

Let M be a manifold of class C^∞. If $t \in \mathbf{R}$, we denote the open interval $(-\infty,t)$ by I_t.

Lemma 5.1. *Let $-\infty < a < b < +\infty$ be real numbers and $G \in \mathsf{D}^{\mathrm{b}}(\mathbf{R})$. Let k be an integer. Assume that*

$$\left(H^j_{[t,+\infty)}(G)\right)_t \cong 0 \quad \text{for all } t\in[a,b)\ \text{, and all } j \leqq k.$$

Then one has

$$H^j((-\infty,b)\,;G) \xrightarrow{\sim} H^j((-\infty,a)\,;G)$$

for all $j<k$, and this morphism is injective for $j=k$.

Proof. We will use Theorem 4.2. Let us put $X=(-\infty,b)$, $F=G|_{I_b}$, and $U_t=I_t$, and verify the hypotheses of the theorem.

(i) $\bigcup_{s<t} I_s = \bigcup_{s<t}(-\infty,s) = (-\infty,t) = I_t$ for all $t\in\mathbf{R}$.

(ii) If $s \leqq t$, then $\overline{I_t - I_s} \cap \operatorname{supp} G = \overline{[s,t)} \cap \operatorname{supp} G = [s,t] \cap \operatorname{supp} G$ is compact.

(iii) For $s \in \mathbf{R}$, we have

$$Z_s = \bigcap_{t>s} \overline{I_t - I_s} = \bigcap_{t>s} \overline{[s,t)} = \bigcap_{t>s} [s,t] = \{s\}\,,$$

and

$$Z_s - I_s = \{s\}\,.$$

By the hypothesis of the lemma, we have

$$H^j_{X-I_s}(F)_s = H^j_{[s,\infty)}(F)_s = 0$$

for $s \in \{s\}$.

(iv) It follows that $Z_s = \{s\} \subset (-\infty, t) = I_t$ if $s < t$.

Then we can apply Theorem 4.2, and we have the isomorphisms for all $t \in [a, b)$ and $j < k$,

$$H^j(I_b; F) = H^j\left(\bigcup_s I_s; F\right) \xrightarrow{\sim} H^j(I_t; F)$$

and the monomorphisms

$$H^k\left(I_b; F\right) = H^k\left(\bigcup_s I_s; F\right) \hookrightarrow H^k(I_t; F).$$

□

We will prove a truncated version of the following proposition.

Proposition 5.2 ([GKS12, Proposition 1.8]). *Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_M)$, and $\psi\colon M \to \mathbf{R}$ be a function of class C^1, proper on* $\operatorname{supp}(F)$. *Let $a < b$ be in $\mathbf{R}$ and assume that $d\psi(x) \notin \mathrm{SS}(F)$ for $a \leqq \psi(x) < b$. For $t \in \mathbf{R}$, set $M_t := \psi^{-1}\left((-\infty, t)\right)$. Then, the restriction morphism*

$$\mathrm{R}\Gamma(M_b; F) \to \mathrm{R}\Gamma(M_a; F)$$

is an isomorphism.

A truncated version of this assertion is the following.

Theorem 5.3. *Let F in $\mathsf{D}^{\mathrm{b}}(\mathbf{k}_M)$, and $\psi\colon M \to \mathbf{R}$ be a real-valued function of class C^1, proper on* $\operatorname{supp}(F)$. *Let $a < b$ be real numbers. We assume $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leqq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}\left((-\infty, t)\right)$. Then, the restriction morphism*

$$H^j(M_b; F) \to H^j(M_a; F)$$

is isomorphic for each $j < k$, and injective for $j = k$.

Proof. First, we have

$$H^j(M_t; F) \cong H^j\,\mathrm{R}\Gamma((-\infty, t)\,; \mathrm{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leqq \psi(x) < b$, and by (3.3), the formula $\mathrm{SS}_k(\mathrm{R}\psi_* F) \subset \psi_\pi \psi_d^{-1}\,\mathrm{SS}_k(F)$ holds. Hence we have

$$\left(H^j_{[\psi(x), +\infty)}(\mathrm{R}\psi_* F)\right)_{\psi(x)} = 0$$

for each $j \leqq k$. Then we can apply Lemma 5.1 and obtain

$$H^j((-\infty, b)\,; \mathrm{R}\psi_* F) \xrightarrow{\sim} H^j((-\infty, a)\,; \mathrm{R}\psi_* F)$$

for all $j < k$ and

$$H^k((-\infty, b)\,; \mathrm{R}\psi_* F) \hookrightarrow H^k((-\infty, a)\,; \mathrm{R}\psi_* F).$$

Then we can conclude that

$$H^j(M_b\,; F) \xrightarrow{\sim} H^j(M_a\,; F)$$

for all $j < k$ and

$$H^k(M_b\,; F) \hookrightarrow H^k(M_a\,; F).$$

□

6 Proof of Lemma 4.4

In this section, we prove Lemma 4.4, which is postponed in the proof of Theorem 4.2.

Proof of Lemma 4.4 First we prove that for $F \in \mathrm{Mod}(\mathbf{k}_X)$, the natural morphism

(6.1)
$$\varinjlim_{t>s} \Gamma(U_t; F) \xrightarrow{\sim} \Gamma(Z_s \cup U_s; F).$$

is isomorphic. Consider the following diagram.

$$\begin{tikzcd}
\varinjlim_{t>s} \Gamma(U_t; F) \arrow[d, "\tilde{\phi}"'] \arrow[dr] & \\
\varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \arrow[r, "\phi"'] \arrow[d] & \Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}) \arrow[d] \\
\varinjlim_{U \supset Z_s} \Gamma(U; F) \arrow[r, "\sim", "\psi"'] & \Gamma(Z_s; F|_{Z_s}).
\end{tikzcd}$$

Let us begin by showing that the morphism

$$\tilde{\phi}\colon \varinjlim_{t>s} \Gamma(U_t; F) \to \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F)$$

is an isomorphism. It is enough to show that $(U_t)_{t>s}$ is a cofinal family of $(U \cup U_s)_{U \supset Z_s}$. Let U be an open neighborhood of Z_s. Then we shall show by contradiction that there exists a real number $t > s$ such that $U_t \subset U \cup U_s$. Let us assume that $U_t - (U \cup U_s) \neq \varnothing$ for any $t > s$. Then define the family of subsets $(A_r)_{s<r<t}$ of $\overline{U_t - U_s}$ by $A_r := U_r - (U \cup U_s)$. This $(A_r)_{s<r<t}$ satisfies the finite intersection property. Indeed, for finite real numbers $s < r_0 < \cdots < r_m < t$, it follows from the assumption for contradiction that

$$\begin{aligned} A_{r_0} \cap \cdots \cap A_{r_m} &= A_{r_0} \\ &= U_{r_0} - (U \cup U_s) \\ &\neq \varnothing. \end{aligned}$$

Therefore, since $\overline{U_t - U_s}$ is compact, we have $\bigcap_{s<r<t} \overline{A_r} \neq \varnothing$. However, we have from the hypothesis

$Z_s \subset U$ that

$$
\begin{aligned}
\varnothing \neq \bigcap_{s<r<t} \overline{A_r} &= \bigcap_{s<r<t} \overline{U_r - (U \cup U_s)} \\
&= \bigcap_{s<r<t} (\overline{U_r - U_s} \cap \overline{U_r - U}) \\
&= \bigcap_{s<r<t} \overline{U_r - U_s} \cap \bigcap_{s<r<t} \overline{U_r - U} \\
&= Z_s \cap \bigcap_{s<r<t} \overline{U_r - U} \\
&= Z_s \cap \bigcap_{s<r<t} \overline{U_r} \cap \overline{X - U} \\
&= Z_s \cap \bigcap_{s<r<t} \overline{U_r} \cap (X - U) \\
&= Z_s \cap (X - U) \cap \bigcap_{s<r<t} \overline{U_r} \\
&= \varnothing.
\end{aligned}
$$

This is a contradiction, and it follows that $\tilde{\phi}$ is an isomorphism.

Since $\tilde{\phi}$ is an isomorphism, it is enough to show that

$$\phi\colon \varinjlim_{U \supset Z_s} \Gamma(U \cup U_s; F) \to \Gamma(Z_s \cup U_s; F)$$

is isomorphic. Since X is Hausdorff and Z_s is compact, the natural morphism

$$\psi\colon \varinjlim_{U \supset Z_s} \Gamma(U; F) \to \Gamma(Z_s; F)$$

is isomorphic by Proposition 2.1. Since U_s is open, we have another isomorphism

$$\psi'\colon \varinjlim_{V \supset U_s} \Gamma(V; F) \to \Gamma(U_s; F).$$

The injectivity of ϕ follows from the definition of a sheaf and the injectivities of ψ and ψ'.

Let us prove that ϕ is surjective. Let $\sigma \in \Gamma(Z_s \cup U_s; F)$ be a section of $F|_{Z_s \cup U_s}$. By the isomorphism ψ above, we can extend $\sigma|_{Z_s}$ to an open neighborhood U of Z_s in X. More precisely, there exists a section τ on U such that $\tau|_{Z_s} = \sigma|_{Z_s}$. On the other hand, it follows from the compatibility of restrictions of a section that $\tau|_{Z_s} = (\tau|_{(Z_s \cup U_s) \cap U})\Big|_{Z_s}$. By the definition of a section on the closed subset Z_s of $(Z_s \cup U_s) \cap U$, we can find an open subset W of $(Z_s \cup U_s) \cap U$ such that

$$\tau|_W = \sigma|_W.$$

By the definition of the induced topology of $(Z_s \cup U_s) \cap U$, there exists an open set $\widetilde{W}$ of X such that

$$W = ((Z_s \cup U_s) \cap U) \cap \widetilde{W} \quad \text{and} \quad \widetilde{W} \subset U.$$

Thus we have

$$U_s \cap W = U_s \cap \widetilde{W}, \quad \sigma|_{U_s \cap \widetilde{W}} = \tau|_{U_s \cap \widetilde{W}}. \tag{6.2}$$

For this $\widetilde{W}$, the pair $\left(U_s, \widetilde{W}\right)$ is an open covering of $\widetilde{W} \cup U_s$. Thus by (6.2), we can glue $\sigma|_{U_s}$ and $\tau|_{\widetilde{W}}$ together and get a section $\widetilde{\sigma}$ such that

$$\widetilde{\sigma}|_{Z_s \cup U_s} = \sigma, \quad \widetilde{\sigma}|_{\widetilde{W}} = \tau|_{\widetilde{W}}.$$

This proves the surjectivity of ϕ.

Next let $F \in \mathsf{D}^+(\mathbf{k}_X)$. Let $I^\bullet$ be a complex of flabby sheaves quasi-isomorphic to F. Note that if S is a subspace of a Hausdorff space such that all the open subsets are paracompact, the restrictions of a flabby sheaf F, denoting by $F|_S$, is also flabby. Then it follows from (6.1) that

$$\begin{aligned}
\varinjlim_{t>s} H^j(U_t; F) &\cong \varinjlim_{t>s} H^j\Gamma(U_t; I^\bullet) \\
&\cong H^j\left(\varinjlim_{t>s} \Gamma(U_t; I^\bullet)\right) \\
&\cong H^j\Gamma(Z_s \cup U_s; I^\bullet) \\
&\cong H^j(Z_s \cup U_s; F).
\end{aligned}$$

This completes the proof. □

Acknowledgement

The author would like to thank Naofumi Honda, his supervisor for patient guidance. The author also thanks Ryosuke Sakamoto, Kango Ito, Yoshiaki Yamamura for fruitful discussions. In particular, Sakamoto gave him several ideas that is contained in this paper. We would like to thank Yuichi Ike for helpful discussions.

References

[FM07] Teresa Monteiro Fernandes, Ana Rita Martins. *Truncated Microsupport and Hyperbolic Inequalities*, Nagoya Mathematical Journal, 2007, Vol. 185, pp.63–91.

[GKS12] Stéphane Guillermou, Masaki Kashiwara, Pierre Schapira, *Sheaf Quantization of Hamiltonian Isotopies and Applications to Nondisplaceability Problems*, Duke. Math. J. 161, No. 2, pp.201–245, 2012.

[KFS03a] Masaki Kashiwara, Teresa Monteiro Fernandes, Pierre Schapira, *Truncated Microsupport and Holomorphic Solutions of D-modules*, Ann. Scient. Éc. Norm. Sup., série 4, tome 36, pp.583–399, 2003.

[KFS03b] Masaki Kashiwara, Teresa Monteiro Fernandes, Pierre Schapira, *Involutivity of Truncated microsupports*, Bull. Soc. math. France, 131 (2), 2003, pp.259–266.

[KS90] Masaki Kashiwara, Pierre Schapira, *Sheaves on Manifolds*, Grundlehren der Mathematischen Wissenschaften, 292, Springer, 1990.